

Smart Spot Application Note

AN04: Installation, maintenance and diagnosis
guide
v1.0



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1. Introduction

This document is a technical guide developed to assist users with the installation, maintenance and troubleshooting of issues related to the Smart Spot device. Should you require any further assistance, please contact our technical support team via email at support@libelium.com.

1.1. General recommendations

- Despite the Smart Spot's high resistance, it is advisable to handle it with caution to ensure its longevity.
- Handle the Smart Spot node with care, ensure that it is not dropped or subjected to undue force.
- Note that it is important to avoid placing the devices in areas where high temperatures may occur, as this may cause damage to the electronic components.
- It is advisable to avoid environments that are characterised by aggressive conditions, such as the presence of corrosive gases, animals, excessive or condensing humidity, or atmospheres with salt suspended in the air.
- To prevent damage to the electronic components, ensure that the Smart Spot is stored in a clean and dust-free environment.
- It is recommended that no paint of any type be used on the device, as this may have a detrimental effect on the operation of the measurements, connections and locking mechanisms.
- Ordinary personnel should not have access to the device (both externally and internally).
- Access to the device should be restricted to expert operators for maintenance or calibration purposes.
- Note that this equipment is not suitable for use in locations where children are likely to be present.
- Note that the antennas should not be forced onto their connectors, as this may cause damage to the connectors.
- Ensure that extension probes are connected to their designated sockets.
- This product can be mounted on a wall, a pole or a street light.
- The recommended installation height is between 3 and 4.5 metres, in order to meet various requirements (vandalism prevention, noise position, etc.).

1.2. Smart Spot physical interface

1.2.1. Common interface

The majority of physical interface components are consistent across models. These include:

- 'Air Quality Extension' socket: This socket allows users to seamlessly communicate the Smart Spot with the Air Quality Extension in 'Air' and 'Air LoRaWAN' models.
- 'Power supply' socket: This socket provides energy to the Smart Spot through any of the available power supply options.
- 'Power Button': The device is powered using this button, which incorporates an LED to indicate the current status.
- 'WiFi/BLE, GNSS and cellular antenna': This is used for the main ways of communicating on the Smart Spot.

- 'External SIM Extension': This optional extension provides users with the ability to set up their own SIM card in the Smart Spot. For more information about the usage of this extension, please refer to the 'Technical Guide' document.

The remaining two interfaces are designed for use with sensor extensions or additional antennas, such as the LoRaWAN. Should you require further information on the installation of external sensor extensions in the sockets, please refer to the dedicated '[Extension sensor sockets](#)' section.

1.1.1.1. On/Off button

The On/Off button is used to switch the unit on and off. Pressing the button will initiate the boot sequence. After a brief period, a blue LED will illuminate to indicate that the device is operational and ready for use. During sleep phases in 'LoRaWAN Sleeper' and 'M2M Sleeper' device modes, the LED will remain illuminated, though at a low intensity.

1.1.1.2. 'Air'/'Crowd' model interface

The 'Air' and 'Crowd' models have the same arrangement of sockets on the unit, as shown in the next image. The key distinction is the usage exclusion, but present, of the 'Air Quality Extension' socket in the 'Crowd Model'.



IMPORTANT: It is imperative to exercise caution when connecting sensors to sockets, as these sockets are utilised for various sensor communication technologies. Failure to ensure the correct socket is used for a sensor can result in damage to the sensor.

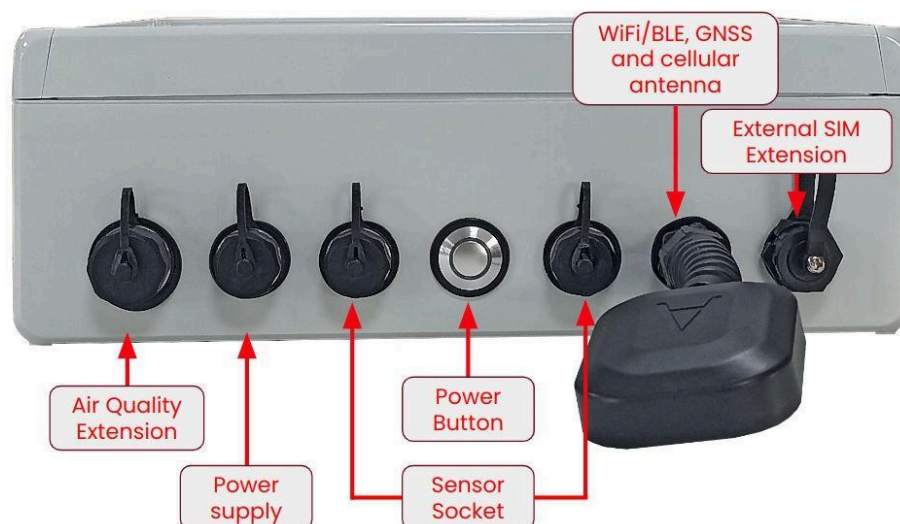


Figure: Smart Spot Crowd and Air socket description 'Air LoRaWAN' model interface

1.1.1.3. 'Air LoRaWAN' model interface

The 'Air LoRaWAN' models closely resemble the 'Air' and 'Crowd' models, but utilise the 'Sensor Socket' in closer proximity to the 'WiFi/BLE, GNSS and Cellular Antenna' to position the LoRaWAN antenna.

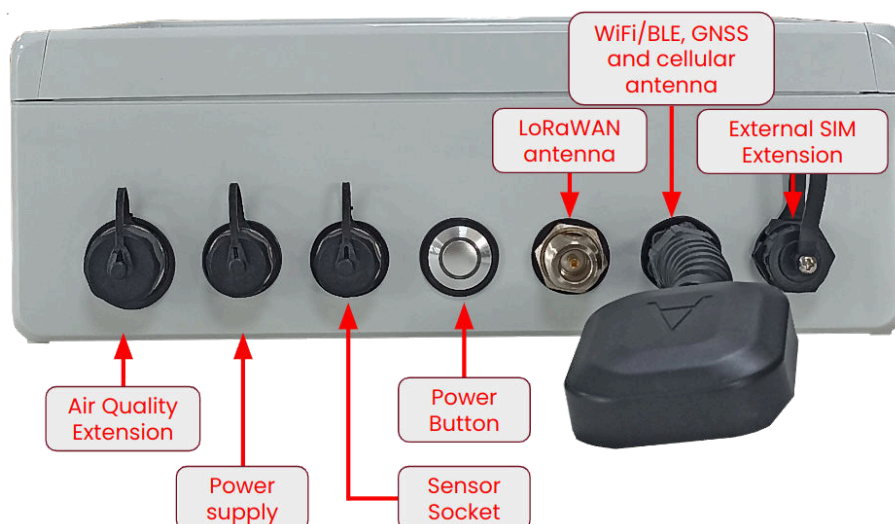


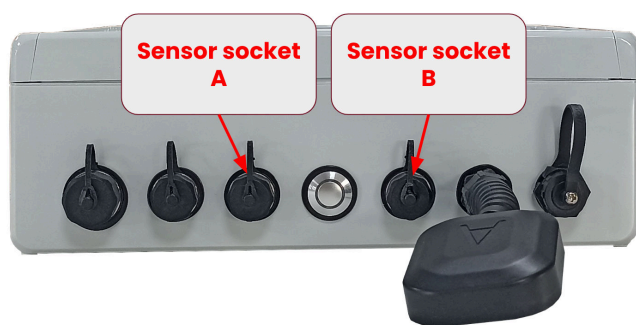
Figure: Smart Spot Air LoRaWAN socket description

1.1.2. Extension sensor sockets

Depending on the model, different sensor sockets arrangements can be found on the core housing. The 'Crowd' and 'Air' models allow the connection of two different external sensor extensions, while 'Air LoRaWAN' occupies one of the sensor sockets with the LoRaWAN antenna.

The following image illustrates the position and availability of the external sensor extensions sockets (A and B) depending on the device model.

Should you require further details regarding the extensions not related with installation, please refer to the dedicated ['Technical Guide'](#) document.



'Crowd' and 'Air' models sensor sockets



'Air LoRaWAN' sensor sockets

1.1.2.1. External sensor extensions for 'Crowd' and 'Air'



IMPORTANT: Please note that when connecting external sensor extensions, caution should be exercised as the sockets are not hardware compatible. This may result in irreparable damage to the sensors.

The following list provides a detailed overview of the external sensor extension's position for each sensor socket.

Sensor socket	Reference	Name
A	SPT-SEN-TH	Temperature and Humidity sensor
	SPT-CM-SEN-NLS2	Noise Level sensor (Only applicable for 'Crowd' models)
B	SPT-SEN-DVP2	Davis Pro 2 sensor
	SPT-SEN-DVPP2	Davis Pro 2 Plus sensor
	SPT-SEN-AWD	Wind anemometer & direction sensor
	SPT-SEN-AWDTHP	Wind speed/direction, T/H and Pressure sensor

1.1.2.2. External sensor extensions for 'Air LoRaWAN'

The following list provides a detailed overview of the external sensor extension's position for each sensor socket.

Senor socket	Reference	Name
A	SPT-SEN-TH	Temperature and Humidity sensor

2. Installation and maintenance

2.1. Power supply

Regardless of the model selected, the power supply is connected to the Smart Spot via a 2-pin connector, located as previously illustrated in the 'Smart Spot Core socket description' section.

2.1.1. Indoor power supply

- **Step 1:** Ensure that the 2-pin cable is connected to the Smart Spot using the socket provided, as outlined in the ['Smart Spot physical interface'](#) section.
- **Step 2:** Connect the power supply to the mains plug.



Figure: Power supply installation

2.1.2. Outdoor power supply

2.1.2.1. Connection

- **Step 1:** Connect the L/N wires to the fuse (the markings on the fuse indicate where to connect them).
- **Step 2:** Please ensure that the 2-pin cable is connected to the transformer.
 - Ensure that the blue cable is connected to the -V terminal, which corresponds to the N line.
 - Ensure that the brown cable is connected to the positive voltage (L line).
- **Step 3:** Ensure that the 2-pin cable is connected to the Smart Spot using the socket provided, as outlined in the ['Smart Spot physical interface'](#) section.

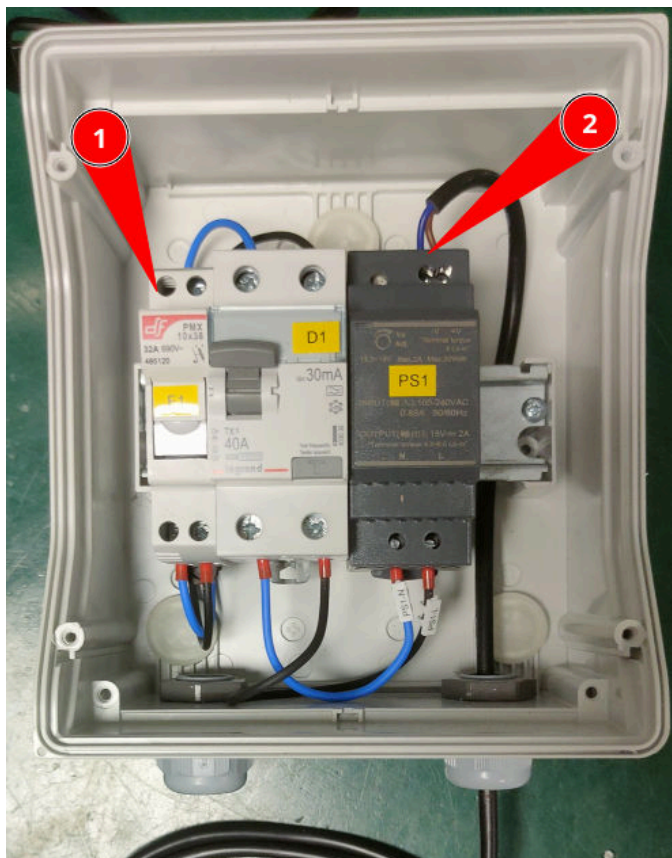


Figure: Outdoor power supply connections: 1) L/N input; 2) 15VDC output

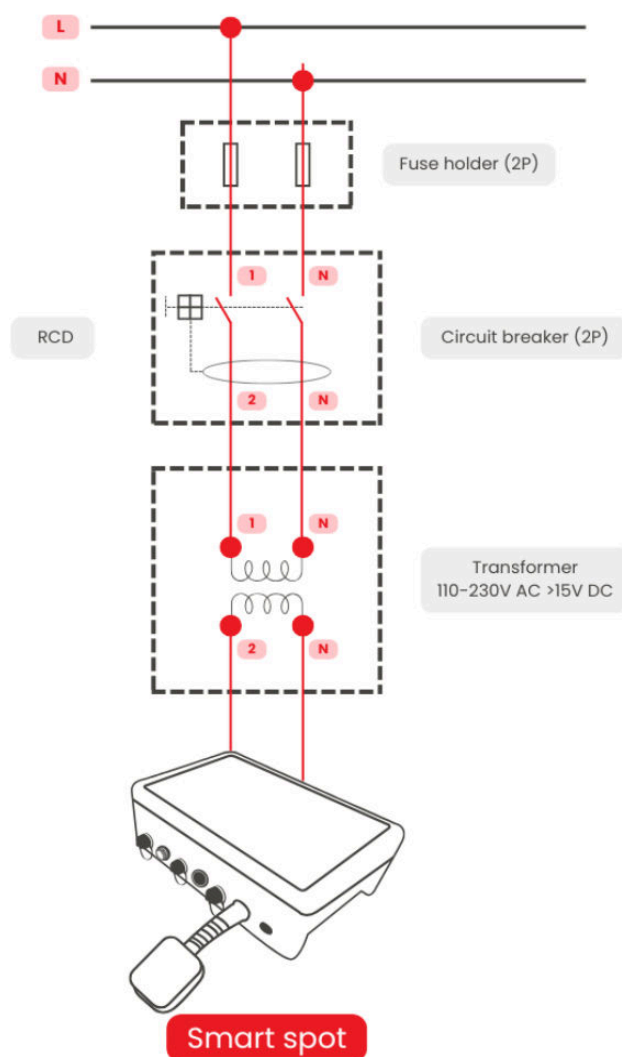


Figure: Electrical schema for outdoor power supply

2.1.2.2. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the outdoor panel is aligned and firmly attached. Please make any necessary adjustments.
- Ensure that the box integrity is correct to prevent water from entering the electrical components.
- Check the cables for any damage.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

2.1.3. Solar Panel

2.1.3.1. Important considerations

The optimal orientation of solar panels is determined by aligning them with the equator (south in the northern hemisphere and north in the southern hemisphere) and adjusting the tilt angle according to the latitude of the installation site.

- Northern hemisphere: It is recommended that solar panels be positioned to face south.

- Southern hemisphere: It is recommended that solar panels be positioned to face north.

The tilt angle of a solar panel is determined by the latitude of the installation site. It is generally considered good practice to use a tilt angle that is equal to the latitude of the site. For example, if the installation is in a location with a latitude of 30 degrees, the panels should be tilted at 30 degrees to the horizontal plane.

There are several other important considerations to take into account when installing the solar panel.

- **Shading:** Please ensure that the panels are not shaded during peak sunlight hours. It is important to take care to avoid shading caused by trees, buildings or other obstructions.
- **Ventilation:** It is also important to leave enough space between the panels and the mounting surface to allow for good ventilation, which helps prevent overheating and improves panel efficiency.

If feasible, it is advisable to use the solar panel to shield the Smart Spot core housing from direct sunlight, thereby preventing the system from reaching excessive temperatures. In the context of air quality models, it is advisable to safeguard the air quality case in order to ensure the system's temperature remains more stable instead of the core housing.

2.1.3.2. Wall mounting

- **Step 1:** Determine the correct position of the panel based on the considerations.
- **Step 2:** Mark the holes by placing the solar panel support in the desired position.
- **Step 3:** Drill the wall and insert the corresponding plugs.
- **Step 4:** Attach the solar panel to the bracket (1).
- **Step 5:** Screw the support to the wall.
- **Step 6:** Ensure that the extension is correctly and securely attached.
- **Step 7:** Ensure that the solar panel cable is connected to the Smart Spot using the socket provided, as outlined in the '[Smart Spot physical interface](#)' section.

2.1.3.3. Pole mounting

- **Step 1:** Determine the correct position of the panel based on the considerations.
- **Step 2:** Attach the solar panel to the bracket (1).
- **Step 3:** Fasten the bracket to the pole using clamps. (2)
- **Step 4:** Ensure that the extension is correctly and securely attached.
- **Step 5:** Ensure that the solar panel cable is connected to the Smart Spot using the socket provided, as outlined in the '[Smart Spot physical interface](#)' section.

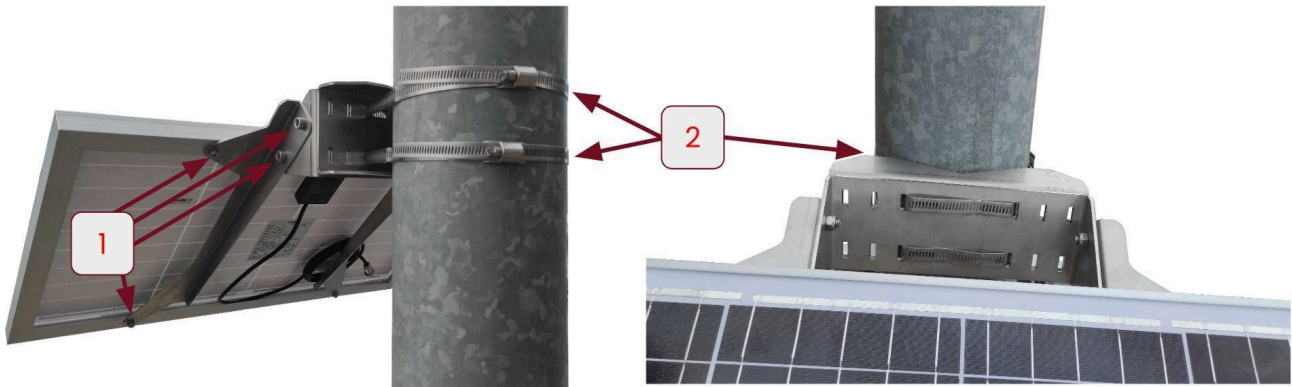


Figure: Solar panel pole installation guide

2.1.3.4. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the solar panel is aligned and firmly attached. Please make any necessary adjustments.
- Check for any damage to the solar panel, as this may reduce its capacity.
- Check the cables for any damage.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

2.2. Smart Spot

Utilities for both wall and pole mounting are provided with the Smart Spot such as illustrated on the next figures.



To account for the wide range of scenarios and customers (poles, street lights of different diameters), endless clamps are offered as a separate item, enabling cost savings for customers.



When using endless clamps, please ensure that the excess is cut off after tightening.

2.2.1. Smart Spot Core housing

The following accessories are included for installation in both pole and wall settings, as illustrated in the following image.



Wall mounting accessories



Pole mounting accessories

Figure: Wall and pole mounting accessories for Smart Spot core housing

2.2.1.1. Wall mounting

- **Step 1:** Ensure that the wall mounting accessories are placed in the core housing.(1).
- **Step 1:** Mark the holes by placing the Smart Spot in the desired position.
- **Step 2:** Drill the wall and insert the corresponding plugs.
- **Step 4:** Screw the core housing to the wall.
- **Step 5:** Ensure that the core housing is correctly and securely attached.

2.2.1.2. Pole mounting

- **Step 1:** Ensure that the pole mounting accessories are placed in the core housing.(1).
- **Step 2:** Use the accessory to pass the end-less clamp and attach the device to the pole (2).

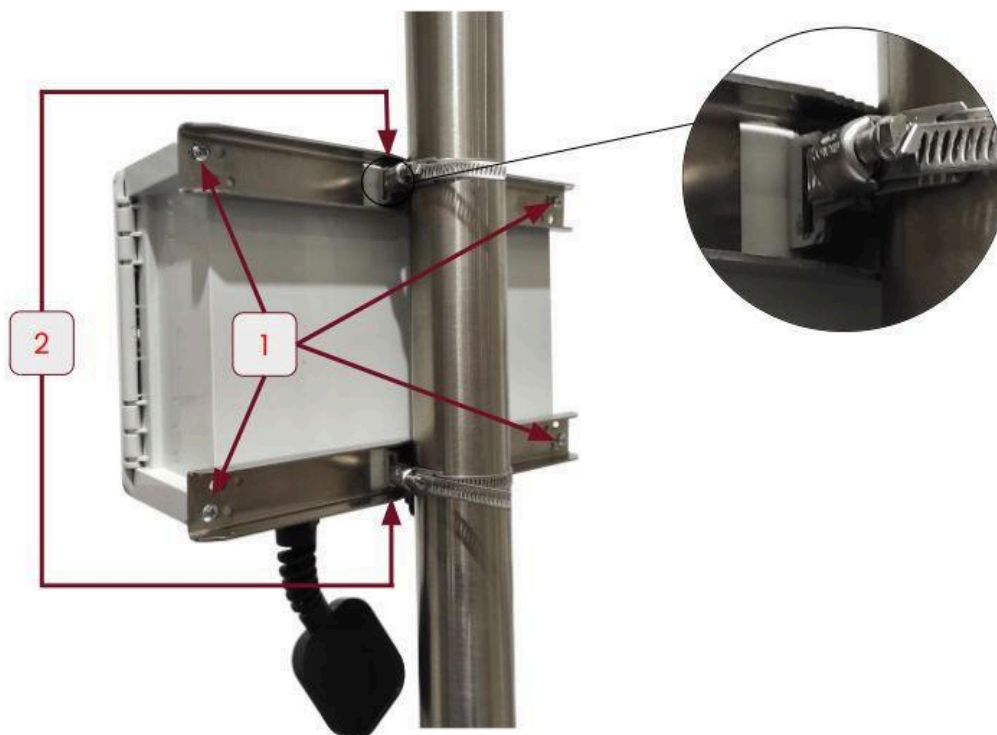


Figure: Steps to install the Core in Pole

2.2.1.3. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the device is aligned and firmly attached. Please make any necessary adjustments.
- Check for general damage to the device, such as issues with encapsulation integrity, antenna integrity, etc.
- Check the cables for any damage.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

2.2.2. Air Quality Extension



Please note that this section is applicable only to the 'Air' and 'Air LoRaWAN' models.

The following accessories are included for installation in both pole and wall settings, as illustrated in the following image.



Figure: Wall and pole mounting accessories for Smart Spot Air Quality Extension housing

2.2.2.1. Important considerations

It is recommended that the units be installed in open and clean areas, but accessible to facilitate maintenance. It is important to avoid proximity to transformers and antennas, as these can generate strong electromagnetic fields that have the potential to interfere with gas sensors. The device must not be installed in areas where smoking is permitted or where smoke extractors are in place.

Prior to installation, it is advisable to consider the type of air quality measurement that is required. There are two main types of air quality measurement:

- Emission (direct pollution): The device must be placed in close proximity to the source of pollution, such as near a road. For example, it should be positioned at a distance of 5 metres from the source.
- Inmission (background pollution): The device must be installed in an area where the pollution is dispersed, such as in open spaces at a distance of 50 metres from the pollution source (parks, gardens, etc.).

Should the 'Air Quality extension' include a sonometer sensor ((SPT-AQ-SEN-NLS2), In accordance with the ISO 19962:2009 standard, the unit **should be installed at a distance of 2 metres from a wall. Avoid installing the product on walls wherever possible.**

If this is not feasible, the Smart Spot allows you to configure a correction parameter for wall installation (-3dB), as described in the standard, to avoid reflected noise. Technical Guide outlines the steps to configure this parameter.

2.2.2.2. Wall mounting

- **Step 1:** Place the mounting accessories in the Extension housing (1)
- **Step 1:** Mark the holes by placing the Smart Spot in the desired position.
- **Step 2:** Drill the wall and insert the corresponding plugs.
- **Step 4:** Screw the Extension housing to the wall.
- **Step 5:** Ensure that the Extension housing is correctly and securely attached.
- **Step 6:** Ensure that the cable 'Air Quality Extension' cable is connected to the Smart Spot using the socket provided, as outlined in the '[Smart Spot physical interface](#)' section.

2.2.2.3. Pole mounting

- **Step 1:** Place the mounting accessories in the Extension housing (1)
- **Step 2:** Use the accessory to pass the end-less clamp and attach the device to the pole (2)

- **Step 3:** Ensure that the cable 'Air Quality Extension' cable is connected to the Smart Spot using the socket provided, as outlined in the '[Smart Spot physical interface](#)' section.

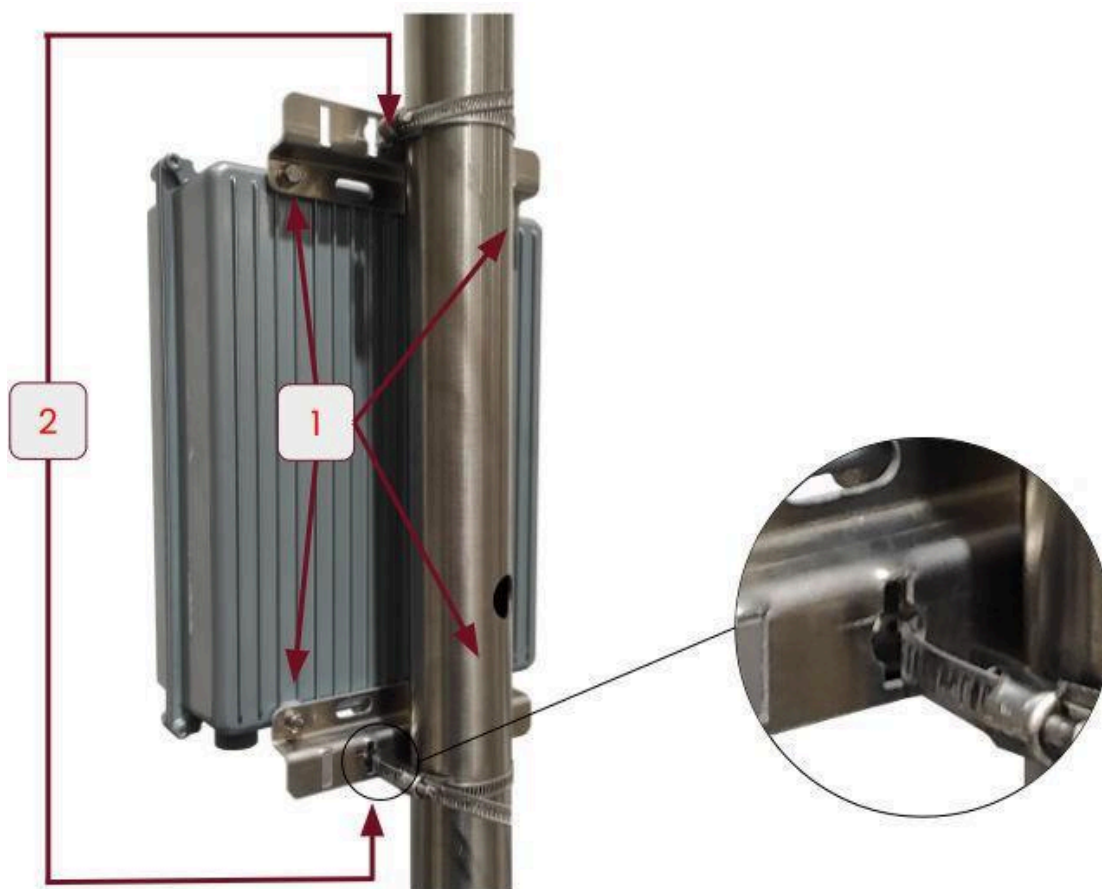


Figure: Steps to install the Air Quality Extension in Pole

2.2.2.4. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the device is aligned and firmly attached. Please make any necessary adjustments.
- Check for general damage to the device, such as issues with encapsulation integrity, air inlets/outlets, etc.
- Check the cables for any damage.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.
- Furthermore, **it is essential to carry out bi-annual maintenance on the air quality sensors embedded in the encapsulation in order to ensure their continued functionality and quality.** Please refer to the '[AN132: Air Quality Sensors Maintenance](#)' document for full details of the maintenance procedures for the following sensors:
 - Electrochemical sensors: NO₂, O₃, CO, SO₂, NO, H₂S and NH₃
 - Particulate Matter
 - VOC
- Should the extension include the Noise Level Extension (SPT-AQ-SEN-NLS2), users are advised to **recalibrate the sonometer during the yearly general maintenance**, in order to verify and ensure the instrument's accuracy. For detailed instructions, please refer to '[Appendix I](#)'.

2.3. Sensor extensions



When using endless clamps, please ensure that the excess is cut off after tightening.

2.3.1. Temperature and humidity sensor

This section outlines the installation procedures for the temperature and humidity sensor (SPT-SEN-TH).

2.3.1.1. Wall mounting

- **Step 1:** Mark the holes by placing the bracket in the desired position.
- **Step 2:** Drill the wall and insert the corresponding plugs.
- **Step 3:** Screw the support to the wall.
- **Step 4:** Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws.
- **Step 5:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. **(Socket A)**

2.3.1.2. Pole mounting

- **Step 1:** Pass the cable through the bracket (1).
- **Step 2:** Fasten the sensor to the bracket using the wing nuts (2).
- **Step 3:** Pass the end-less clamp through the support and attach the sensor to the pole (3).
- **Step 4:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. **(Socket A)**

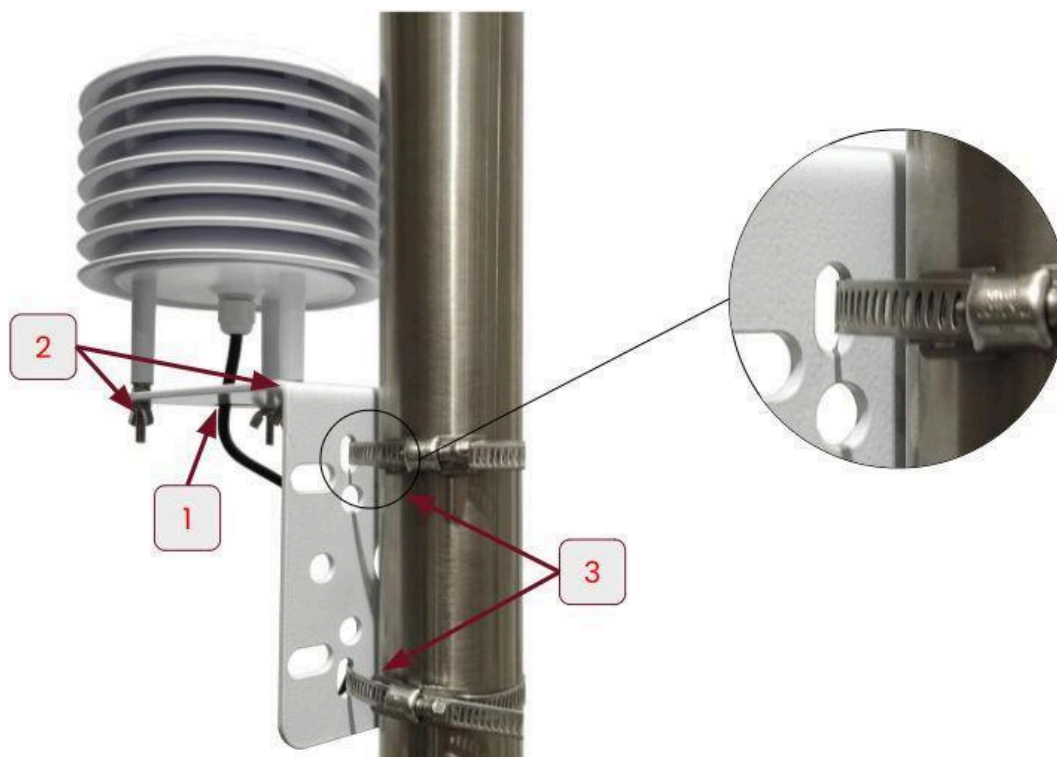


Figure: Steps to install the Temperature and humidity sensor in Pole

2.3.1.3. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the sensor is aligned and firmly attached. Please make any necessary adjustments.
- Check the cables for any damage.
- Ensure the solar shield is properly oriented to minimise direct sunlight on the sensor.
- Check for any insects, dirt or other contaminants inside the solar shield, as these could potentially affect sensor readings.
- Use a portable temperature and humidity meter to take comparative readings. Please ensure the sensor is functioning within the specified accuracy range.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

2.3.2. Wind anemometer and direction sensor

This section outlines the installation procedures for the wind anemometer and direction sensor (SPT-SEN-AWD).

There are four methods for installing the sensor, as illustrated in the figures below.

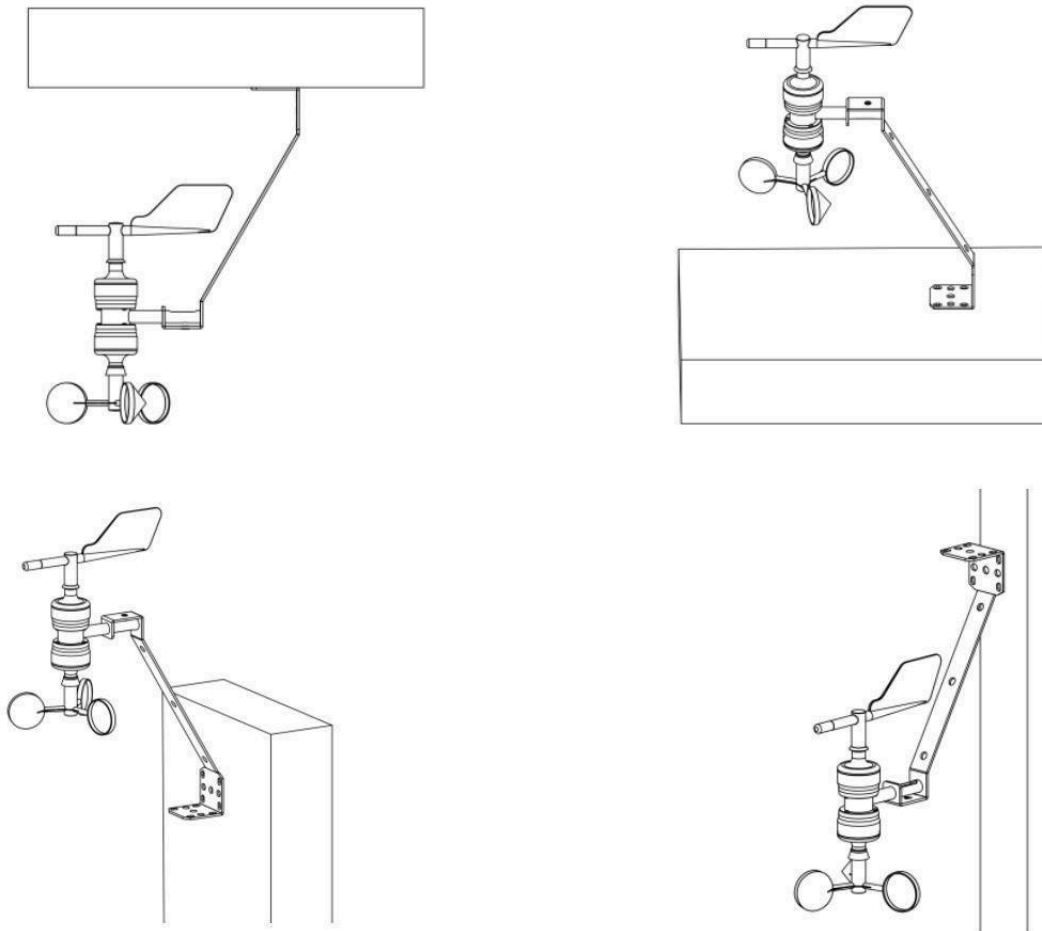


Figure: Installation of wind anemometers and direction sensors

2.3.2.1. Important considerations

Please note that the **wind speed sensor must be installed facing upwards**.

Please note that the sensor must be installed in a specific orientation in order for the wind direction sensor to function correctly. Use the 'S' (South) mark on the sensor to ensure correct installation. **The 'S' (South) mark should point towards the true south**. True south should not be confused with magnetic south. Magnetic variation around the world can be quite large.

Please consider the environment when positioning the sensor, to **avoid placing it under, near or between objects that may interfere with the rain, solar and wind measurements**.

2.3.2.2. Wall mounting

- **Step 1:** Mark the holes by placing the bracket in the desired position
- **Step 2:** Drill the wall and insert the corresponding plugs
- **Step 3:** Screw the support to the wall
- **Step 4:** Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws.
- **Step 5:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the ['Extension sensor sockets'](#) section. **(Socket B)**

2.3.2.3. Pole mounting

- **Step 1:** Insert the clamps in the available bracket positions
- **Step 2:** Place the support in the desired position
- **Step 3:** Close the clamps by firmly tightening the screws.
- **Step 4:** Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws.
- **Step 5:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. **(Socket B)**

2.3.2.4. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the sensor is aligned and firmly attached. Please make any necessary adjustments.
- Check the cables for any damage.
- Ensure the wind sensor indication is correctly oriented to point south.
- Ensure it is freely rotating and not obstructed by any external objects.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.
- Inspect and replace any worn or damaged components such as cups, bearings, wind vane elements, or any seals.

2.3.3. Wind speed/direction, temperature, humidity and pressure sensor

This section outlines the installation procedures for the wind speed/direction, temperature, humidity and pressure sensor (SPT-SEN-AWDTHP).

2.3.3.1. Important considerations

Please note that the sensor must be installed in a specific orientation in order for the wind direction sensor to function correctly. Use the 'arrow' mark (position 1, as shown in the next image) on the wind direction sensor to ensure correct installation. **The 'arrow' mark must point to the true north.** True north should not be confused with magnetic north. Magnetic variation around the world can be quite large.

Please consider the environment when positioning the sensor, to **avoid placing it under, near or between objects that may interfere with the rain, solar and wind measurements.**

2.3.3.2. Installation at the top of the pole

The sensor is designed to be fastened at the top of a pole with a maximum external diameter of 50 mm and a minimum internal diameter of 36 mm using the included screws, or by positioning the flanges in the position 5 illustrated in the next image.

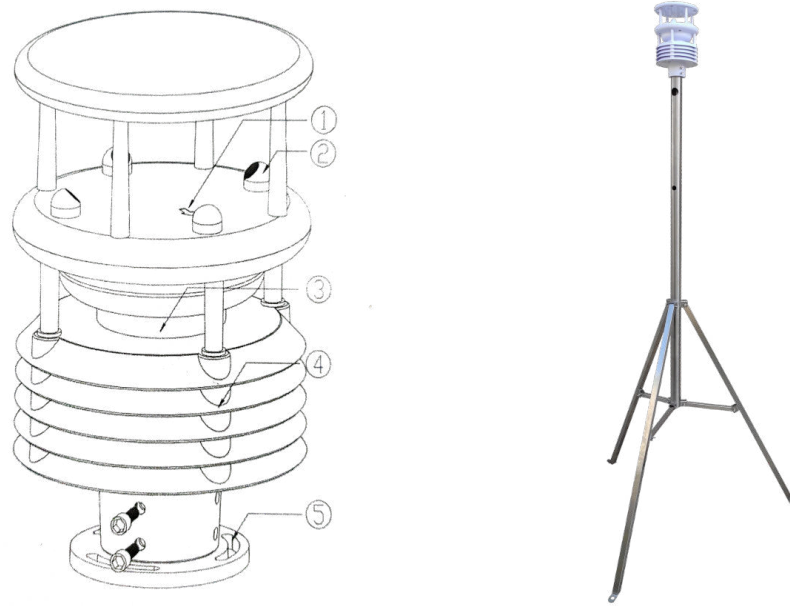


Figure: Installation of the AWDTHP sensors and example of a tripod situation

2.3.3.3. Wall mounting



The sensor is designed to be fastened at the top of a pole as outlined in [‘Installation at the top of the pole’](#) section. Should you require a metallic holder for the sensor, this is available in our catalog. The sensor can thus be attached to a pole with a diameter of up to 210 mm, or to a wall.

- **Step 1:** Mark the four holes by placing the bracket in the desired position (1)
- **Step 2:** Drill the wall and insert the corresponding plugs
- **Step 3:** Screw the support to the wall
- **Step 4:** Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws.
- **Step 5:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the [‘Extension sensor sockets’](#) section. **(Socket B)**

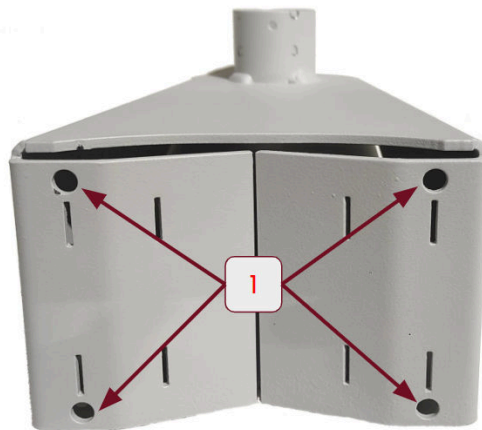


Figure: Holes for straightforward wall installation.

2.3.3.4. Pole mounting



The sensor is designed to be fastened at the top of a pole as outlined in '[Installation at the top of the pole](#)' section. Should you require a metallic holder for the sensor, this is available in our catalog. The sensor can thus be attached to a pole with a diameter of up to 210 mm, or to a wall.

- **Step 1:** Insert the clamps in the available bracket positions (1)
- **Step 2:** Place the support in the desired position (2)
- **Step 3:** Close the clamps by firmly tightening the screws. (3)
- **Step 4:** Place the sensor on the support and once you are sure of the position, firmly tighten the retaining screws. (4)
- **Step 5:** Ensure that sensor cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. **(Socket B)**

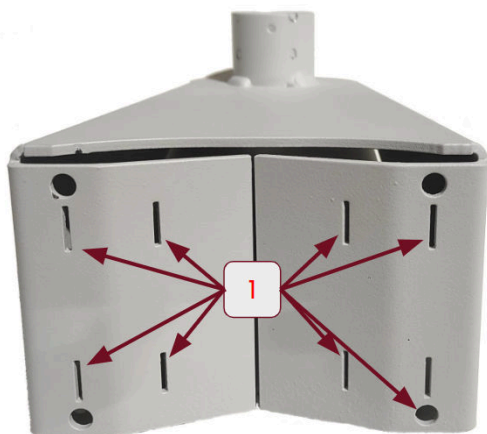


Figure: Available clamp positions in bracket

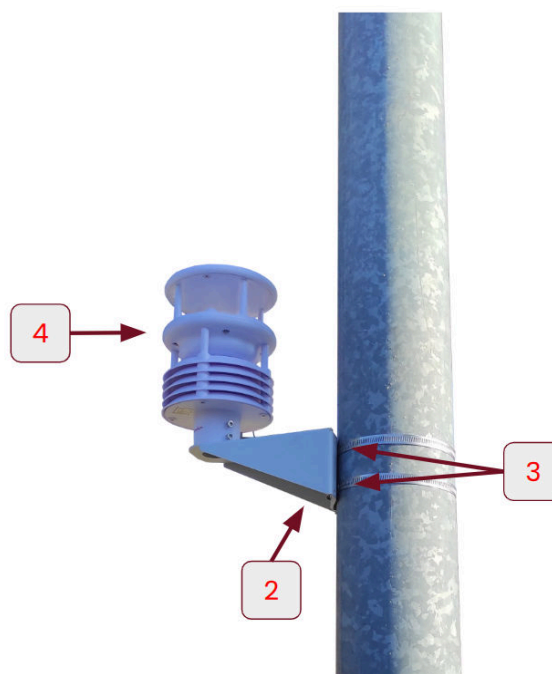


Figure: Positioning and attachment

2.3.3.5. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the sensor is aligned and firmly attached. Please make any necessary adjustments.
- Check the cables for any damage.
- Ensure the wind sensor indication is correctly oriented to point north.
- Inspect and replace any damaged parts.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

2.3.4. Davis Pro2 and Davis Pro2 Plus sensors

This section outlines the installation procedures for the Davis Pro2 (SPT-SEN-DVP2) and Davis Pro2 Plus (SPT-SEN-DVPP2) sensors.



The information in this section has been extracted from the official installation instructions found on the box. If you need more details, please refer to the official installation instructions.

2.3.4.1. Important considerations

Please note that the wind speed and direction sensor must be installed in a specific orientation in order for the wind direction sensor to function correctly. It is **imperative that the sensor arm is oriented towards the true north**. It is important to note that true north should not be confused with magnetic north. Magnetic variation around the world can be quite large.

Please consider the environment when positioning the sensor, to **avoid placing it under, near or between objects that may interfere with the rain, solar and wind measurements**. The following illustration depicts the various scenarios that should be avoided.

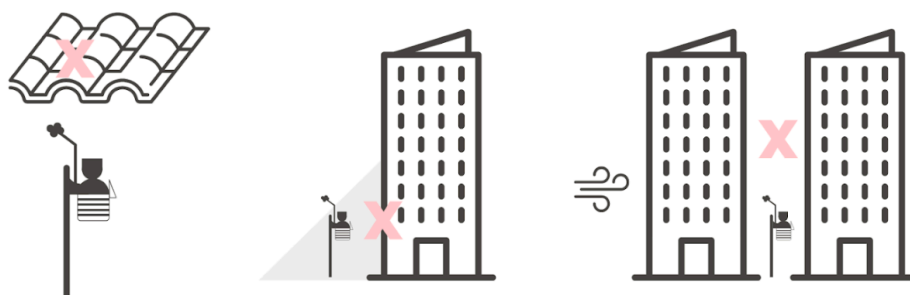


Figure: David Pro 2 installation recommendations

2.3.4.2. Components mounting

Please find below a step-by-step guide to prepare the Davis Pro2 or Davis Pro2 Plus before proceeding with the installation. Once these steps have been completed, you can proceed with the installation.

- **Step 1:** Insert the anemometer arm into the base, sliding the cable through the notch in the base.
- **Step 2:** Line up the small hole in the arm, insert the machine screw through the holes and tighten the hex nut.
- **Step 3:** Push the wind cups up onto the anemometer's stainless steel shaft and use the Allen wrench provided to firmly tighten the set screw on the side of the wind cups.

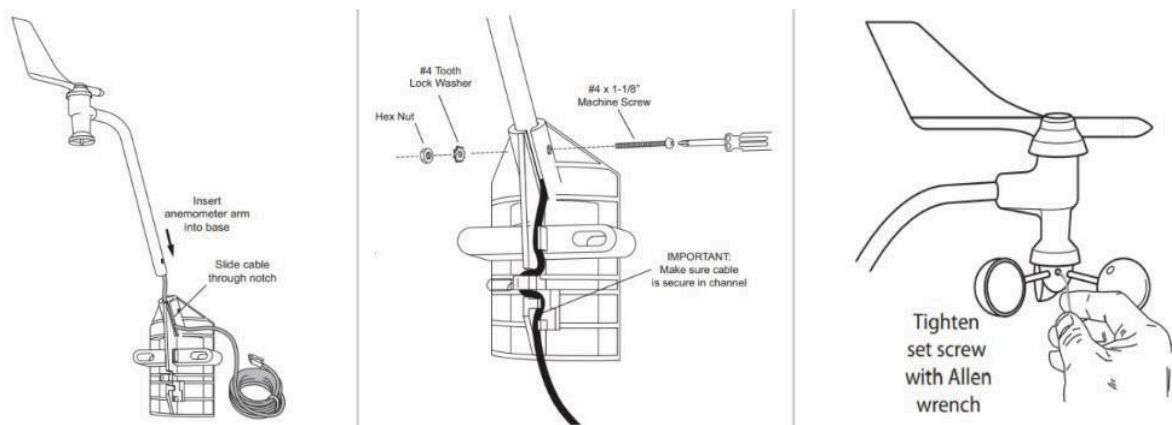


Figure: Davis Pro2 component mounting (Steps 1, 2 and 3)

- **Step 4:** Pull the tab away from the box while sliding the cover up.
- **Step 5:** Pull the foam insert out of the cable access port. Insert the sensor cables and connect them to their specific port. Put back the foam.
- **Step 6:** Twist off the rain collector cone.

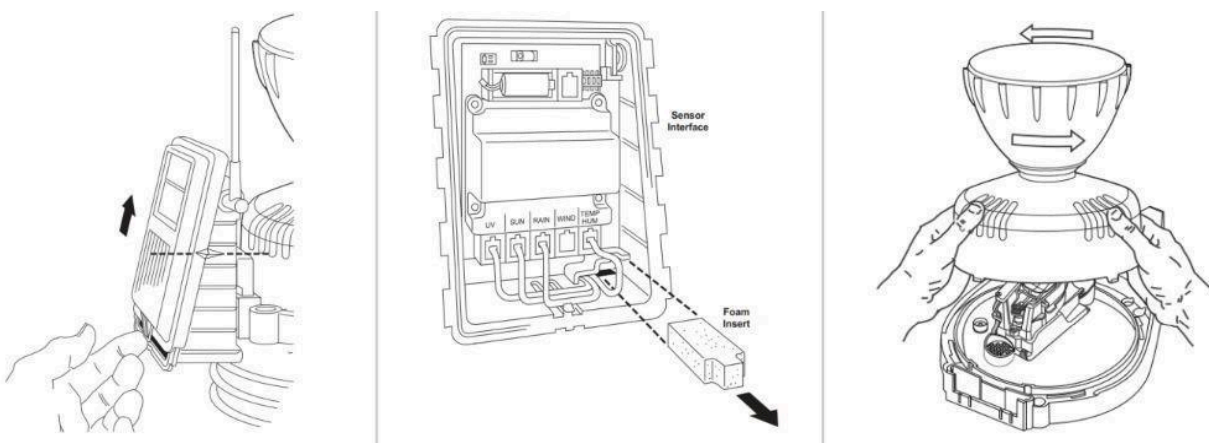


Figure: Davis Pro2 component mounting (Step 4, 5 and 6)

- **Step 7:** Carefully cut and remove the plastic tie that holds the tipping spoon in place during shipping

**IMPORTANT:**

- Ensure that the plastic tie is cut and removed with care in order to allow rain parameters to be measured.

- **Step 8:** Set the cone back on the base, place the debris screen and insert one spike into each socket around the rim of the cone.
- **Step 9:** If includes a solar radiation and UV sensor, use the bubble level on the sensors as a guide to verify that the sensors are level

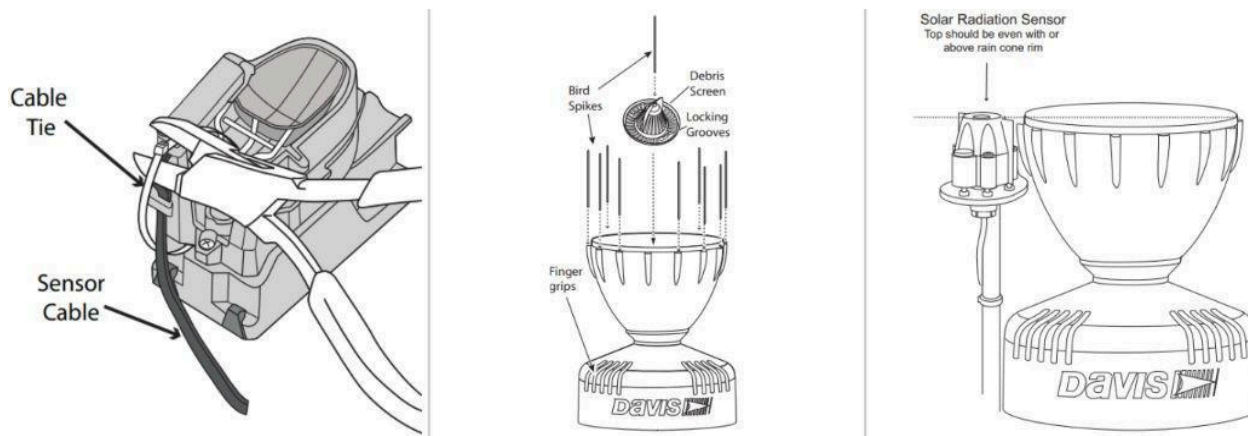


Figure: Davis Pro2 component mounting (Step 7, 8 and 9)

2.3.4.3. Wall mounting

Davis Pro 2:

- **Step 1:** With a 3/16" (5 mm) drill bit, drill two holes approximately 2 1/8" (54 mm) apart. Use a carpenter's level to ensure the holes are level. Use the metal backing plate as a guide when marking the holes.
- **Step 2:** Remove the rain collector cone if it is installed on the Davis Pro2 mounting base.
- **Step 3:** Insert the 1/4" x 3" lag screws through the metal backing plate and the holes in the mounting base into the post. Make sure the Davis Pro2 is level by checking the built-in bubble level.
- **Step 4:** Tighten the lag screws using an adjustable wrench or 7/16" wrench.
- **Step 5:** Re-attach the rain collector cone. Set the cone back on the base so its latches slide downward into the latch openings on the base. Rotate the cone clockwise.
- **Step 6:** Place the debris screen (shown in the illustration on page 1) inside the cone, "feet down" over the funnel hole.
- **Step 7:** Ensure that extension cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. **(Socket B)**

Anemometer:

- **Step 1:** With a 3/16" (5 mm) drill bit, drill two holes approximately 2 1/8" (54 mm) apart. Use a carpenter's level to ensure the holes will be level.
- **Step 2:** Insert the 1/4" x 3" lag screws through the flat washers and the holes in the anemometer mounting base into the post.
- **Step 3:** Tighten the lag screws using an adjustable wrench or 7/16" wrench.

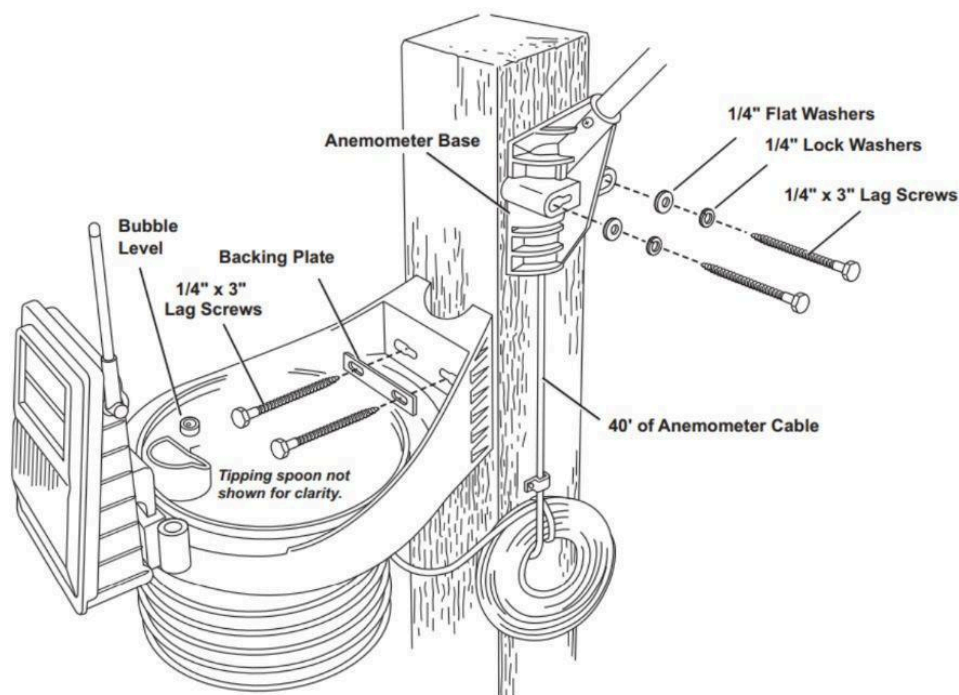


Figure: Davis Pro2 and Davis Pro2 Plus mounting in wall or flat surface

2.3.4.4. Pole mounting



Although the Davis Pro 2 and anemometer arm can be installed separately, the following steps describe mounting the Davis Pro 2 and the anemometer arm together, as illustrated in the figure.

- **Step 1:** Place the U-bolt for the anemometer around the pole so that its round end fits in the top groove of the side of the rain collector side's plastic mounting base. The groove is right above two large holes.
- **Step 2:** While holding the mounting base of the rain collector against the pole, place the two ends of the remaining U-bolt around the pole and through the two holes in the base.
- **Step 3:** Slide the metal backing plate over the bolt ends as they stick out over the rain collector base. Secure the backing plate with a lock washer and hex nut on each of the bolt ends as shown previously.
- **Step 4:** The two ends of the anemometer's U-bolt should now be pointing away from the mounted rain collector side. Slide the anemometer's mounting base over the protruding bolt ends. Place a flat washer, a lock washer and a hex nut on each of the bolt ends as shown above. Do not tighten the nuts yet.
- **Step 5:** Raise the Davis Pro2 unit to the desired height on the pole and swivel it so the anemometer arm is pointing north.
- **Step 7:** Using an adjustable wrench or 7/16" wrench, tighten all four hex nuts until the Davis Pro2 is firmly fastened on the pole.
- **Step 8:** Re-attach the rain collector cone by setting the cone back on the base so its latches slide downward into the latch openings on the base, then rotate the cone clockwise.
- **Step 9:** Place the debris screen inside the cone, "feet down," over the funnel hole. When installing the Davis Pro2 as a single unit, we recommend tucking the coil of anemometer cable between the rain collector cone and the Davis Pro2 base

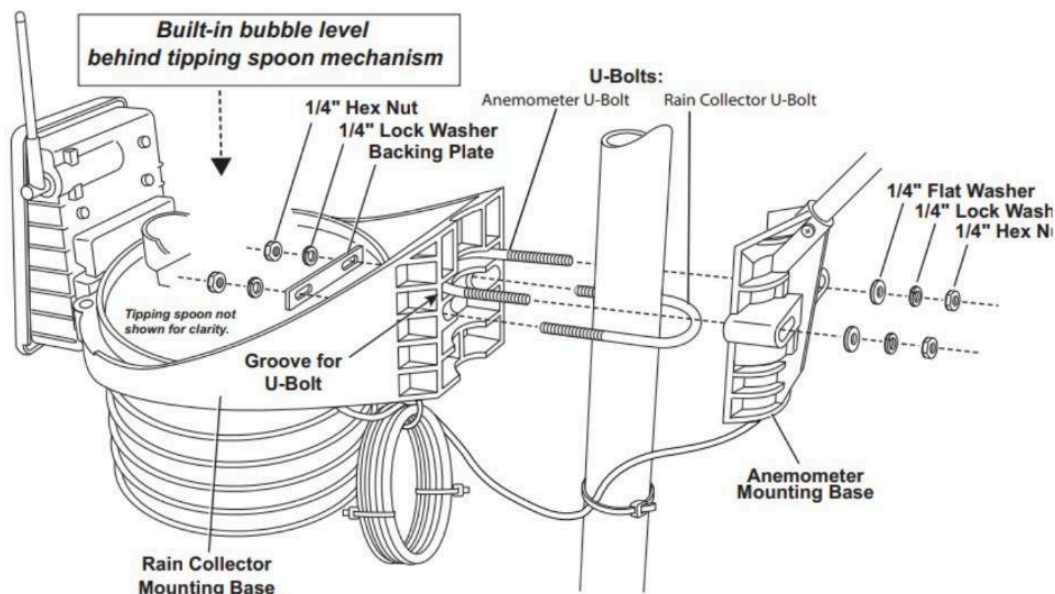


Figure: Davis Pro2 and Davis Pro2 Plus mounting in pole

2.3.4.5. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the sensor is aligned and firmly attached. Please make any necessary adjustments.
- Check the cables for any damage.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.
- Ensure the wind sensor indication is correctly oriented to point north.
- Ensure the rain gauge is balanced and adjust it if necessary.
- Verify that the wind vane rotates freely.
- Make sure the wind vane can rotate without obstruction.
- Remove the debris screen and clean out the debris in the cone. For greatest accuracy, clean your rain collector a few times a year.

2.3.5. Noise Level sensor

This section outlines the installation procedures for the Noise Level sensor (SPT-CM-SEN-NLS2).

2.3.5.1. Important considerations

In accordance with the ISO 19962:2009 standard, the unit **should be installed at a distance of 2 metres from a wall. Avoid installing the product on walls wherever possible.**

If this is not feasible, the Smart Spot allows you to configure a correction parameter for wall installation (-3dB), as described in the standard, to avoid reflected noise. Please refer to '[Appendix II](#)', which details the process for implementing this compensation in relation to the measurements. Each sonometer is calibrated independently, so please do not exchange the microphone between sonometers. For further assistance, please refer to the '[Appendix I](#)' guide, which provides detailed instructions for recalibrating the sonometer.

2.3.5.2. Wall mounting

- **Step 1:** Place the mounting accessories in the extension (1)
- **Step 2:** Mark the holes by placing the Smart Spot in the desired position.
- **Step 3:** Drill the wall and insert the corresponding plugs.
- **Step 4:** Screw the extension to the wall.
- **Step 5:** Ensure that the extension is correctly and securely attached.
- **Step 6:** Ensure that the microphone in the sonometer is connected. (3)
- **Step 7:** Ensure that the cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. (4) **(Socket A)**

2.3.5.3. Pole mounting

- **Step 1:** Place the mounting accessories in the extension. (1)
- **Step 2:** Use the accessory to pass the end-less clamp and attach the device to the pole. (2)
- **Step 3:** Ensure that the extension is correctly and securely attached.
- **Step 4:** Ensure that the microphone in the sonometer is connected. (3)
- **Step 5:** Ensure that the cable 'Air Quality Extension' cable is connected to the Smart Spot using the socket provided, as outlined in the '[Extension sensor sockets](#)' section. (4) **(Socket A)**

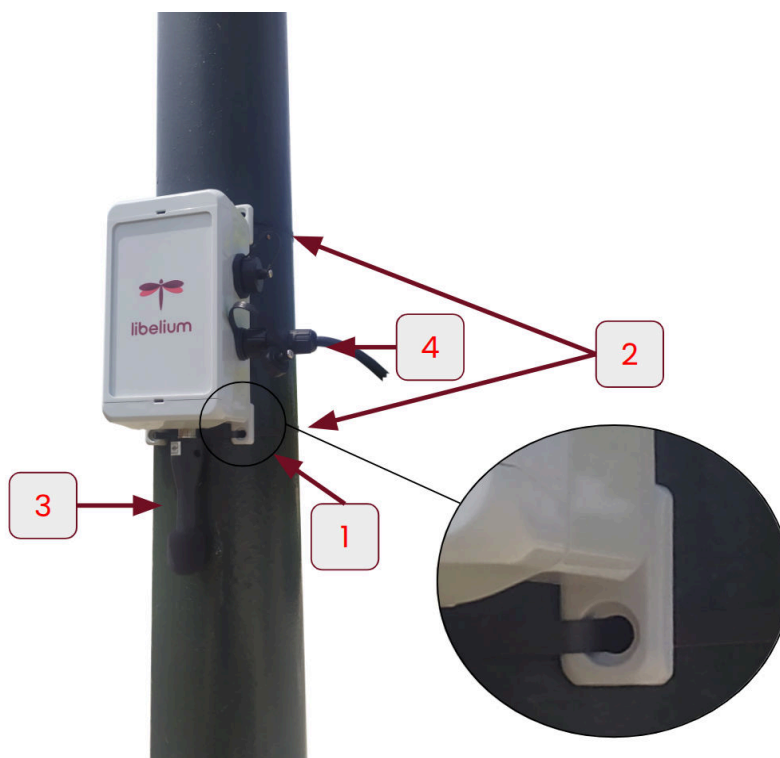


Figure: Noise level sensor mounting steps

2.3.5.4. Maintenance

- It is recommended that an annual maintenance procedure be carried out.
- Verify that the post and/or the sensor is aligned and firmly attached. Please make any necessary adjustments.
- Check the cables for any damage.

- It is recommended that the sonometer be recalibrated at least once during the yearly general maintenance, in order to verify and ensure the instrument's accuracy. For detailed instructions, please refer to '[Appendix I](#)'.
- For optimal cleaning results, use a damp cloth and refrain from using aggressive chemical products.

3. Problem resolution

The present section outlines the steps to follow in order to solve possible problems related with the device functionality.

3.1. Frequently Asked problems and questions

The device does not report any value:

There are a number of potential causes for this issue. In order to assist you in resolving the issue, we recommend commencing the process by following the steps outlined in ["3. Visual inspection"](#).

The device does not connect

There is a possibility that your device is currently without power or is experiencing an issue with its connectivity. We recommend commencing the process by following the steps outlined in ["3. Visual inspection"](#).

How can I check if the device is connected?

The iris360 platform facilitates the viewing of connected devices and offers a range of configuration options. Please refer to section ["4. iris360 connection check"](#) for a comprehensive guide on how to verify the connection status of your device.

The device is connected but does not report any value

There is a possibility that the expansion module or external sensors are not properly connected to the main module. To ensure a smooth and effective process, we recommend that you follow the steps outlined in ["3. Visual inspection"](#) and ["7. Expansion box connection check"](#).

3.2. Relevant checking procedures

3.2.1. Visual inspection

The devices are often installed in public spaces, so it is important that they are installed at an adequate distance from the ground to avoid possible vandalism. The recommended measurement falls between 3.5 and 4.5 metres.

If you encounter any issues with the device's connection, we advise you to first visually verify that the device is still correctly installed in its original location and that all external modules and sensors are in optimal condition.

- **Step 1:** Visual Inspection
 - Please ensure that all enclosures that comprise the device are in their original location.
 - Please ensure that the communication module is fitted with the antenna.
 - Please ensure that the transformer is connected to both the grid and the device.
 - Please ensure that all external sensors (temperature, noise, etc.) are in their original location and in perfect condition.
 - Please ensure that the various modules are connected to each other using the designated communication cable.
 - Please ensure that the blue LED of the power button is switched on (it may be that the device is switched off).



Figure: The power button is illuminated by a blue LED when the unit is switched on.

Should you observe any incident during the visual inspection, please contact technical support by writing to support@libelium.com, indicating the subject line:
Incident Vandalism <Customer_Name> <DEVICE_ID>

If all is in order, please proceed with the steps outlined in the following sections.

3.2.2. iris360 connection check

In order to resolve an issue where you are unable to view any current data on your device, you must first check whether the device is connected to the iris360 platform.

The following steps outline the process to check the last data sent by the device to the iris360 platform.

- **Step 1:** Log in to the Iris360 platform using your user name and password.
- **Step 2:** Navigate to the 'Data Sources' > 'Devices' section.
- **Step 3:** Use the filter box to search for the device(s) you wish to configure if required.
- **Step 4:** Locate the device you wish to check. It's possible to use the filter to facilitate the location of the device in the list.
- **Step 5:** Please verify that the most recent data time falls within the reporting interval. This will help to determine if the device is currently connected to the platform.

If the device is not currently connected, there are two main possibilities: either there is a problem with the power supply or a problem with the connectivity. In order to identify which of these two is the cause of the device disconnection, please follow the instructions in the sections entitled "[5. Power supply check](#)" and "[6. Connectivity check](#)".



Figure: Navigation process through Iris360 to the device's most recent data reporting time.

3.2.3. Power supply check

The following steps outline the process to check the power supply.

- **Step 1:** Please ensure that the transformer is connected to the device.
- It is imperative that the transformer is connected to the main core enclosure via a metal connector at the end of the transformer cable. Please ensure that it is correctly connected, screwed in and free from any looseness.



Figure: Transformer connection with the device

- **Step 2:** Please check that the plug or socket used is functioning correctly.
- It is possible to verify this using a mobile phone charger. In order to complete this task, please disconnect the device's transformer from the socket. Next, connect the phone charger and check that it is charging your mobile device. Should you require the voltage of the socket to be checked, this can be carried out by a qualified professional using a multimeter.

If the socket is not live, please contact your mains network administrator to resolve the issue before proceeding with the following steps.

- **Step 3:** Please reconnect the transformer of the device to the power socket or mains
- **Step 4:** Please ensure that the blue LED of the power button is switched on (it may be that the device is switched off).

Once these steps have been successfully completed, please re-check the connection of the device to the iris360 platform (see section "[4. iris360 connection check](#)"). If the device still does not appear to be connected in some minutes, please continue with the steps described in the following sections.

3.2.4. Device functional check



Perform this step after you have checked the power supply of the device (see section ["5. Power supply check"](#)), once the blue LED of the power button has been illuminated

The subsequent steps detail the procedure for configuring a well-known network for the device, enabling it to connect to the internet network and iris360, thereby verifying its functionality.

- **Step 1:** Please establish a WiFi hotspot with internet access by using a mobile phone. The following credentials should be entered:

- Network name: defaultSSAP
- Password: defaultSSAP1234



- The 'defaultSSAP' is a predefined, well-known network in the device used for setup and checking. It should be noted that this known network can be configured or disabled through iris360. Therefore, it is recommended to check if this step has been performed previously to ensure that the correct credentials have been used, in case they have been modified.
- The generation of a WiFi hotspot is now a standard feature of most smartphones on the market.
- To ensure a seamless internet connection, please ensure you are connected to a 4G network and that the WiFi hotspot is generated.
- It is mandatory to create a WiFi hotspot using the 2.4Ghz band. Please note that SmartSpot is not compatible with the 5Ghz band.
- It has been reported that certain smartphones are experiencing issues with internet connectivity through the WiFi hotspot when a call is in progress. Please refrain from calling during the verification process.

- If possible, please check that the WiFi network used for the check is available and has internet access.
 - Please use an alternative smartphone or computer to check that the WiFi network is available at a distance of less than 5 metres from the device. If you are using a mobile phone for this step, please ensure that network data is deactivated.
 - Once connected, please check that you have internet access by going to www.libelium.com.
- **Step 2:** Please wait between two and five minutes after creating the access point and check if the device is connected to the iris360 platform, as specified in section ["4. iris360 connection check"](#).

If the device is not connecting to the WiFi hotspot, please contact technical support by writing to support@libelium.com, indicating the following in the subject line:
Incident Functional device <Customer_Name> <DEVICE_ID>

3.2.5. Network configuration check



Perform this step after you have checked the power supply of the device (see section ["5. Power supply check"](#)), once the blue LED of the power button has been illuminated

Depending on the type of connectivity used by the device to reach the internet network:

- Please refer to the instructions in section ["6.1. WiFi connectivity check"](#) if the device aims to use WiFi connectivity.
- Please refer to the instructions in section ["6.2. M2M connectivity check"](#) if the device aims to use 2G/3G/4G or NB-IoT connectivity

3.2.5.1. WiFi connectivity check



This section should only be followed if the device in question connects to the iris360 platform via WiFi.

- **Step 1:** Please check that the WiFi network used is available and has internet access.
 - Please use an alternative smartphone or computer to check that the WiFi network is available at a distance of less than 5 metres from the device. If you are using a mobile phone for this step, please ensure that network data is deactivated.
 - Once connected, please check that you have internet access by going to www.libelium.com.

If the network is unavailable or you are unable to access the internet, please contact your provider or manager to resolve the issue.

- **Step 2:** Check that the WiFi network has the required ports available and not captive portals are required.
 - It is imperative that the WiFi network complies with the Smart Spot connectivity requirements, as outlined in the "AN10: Connection Requirements" document.

Should you encounter a network with a captive portal or ports restrictions, please contact your service provider or manager to resolve the issue.

3.2.5.2. M2M connectivity check



This section should only be followed if the device in question connects to the iris360 platform via 2G/3G/4G or NB-IoT.

- **Step 1:** Please check that the SIM card is active.
 - If Libelium is the provider of your SIM card

Please contact our technical support team by emailing support@libelium.com and ensuring to include the subject line.

SIM Card Connectivity Info <Customer_Name> <DEVICE_ID>

- If your carrier has provided the SIM card for the device

Please liaise with your carrier to confirm that the SIM card has been successfully activated and is not subject to a consumption limit that could have resulted in its blocking.



In the event of an external carrier's SIM card being in use, it is possible to check it on a smartphone to verify that it is working properly.

- **Step 2:** Review the SIM card configuration on the device
- Please use the 'Technical Guide' document to verify the configuration of the SIM card. Please be sure to pay particular attention to the APN, as well as any other values that may be required, such as a PIN or username.

3.2.6. Air Quality Expansion enclosure connection check

Should your device be connected to iris360 yet fail to report values related to gas sensors, particulate matter and noise, there may be an issue with the connection between the main and Air Quality expansion enclosures.

- **Step 1:** Please check the connection of the expansion module to the communication module.

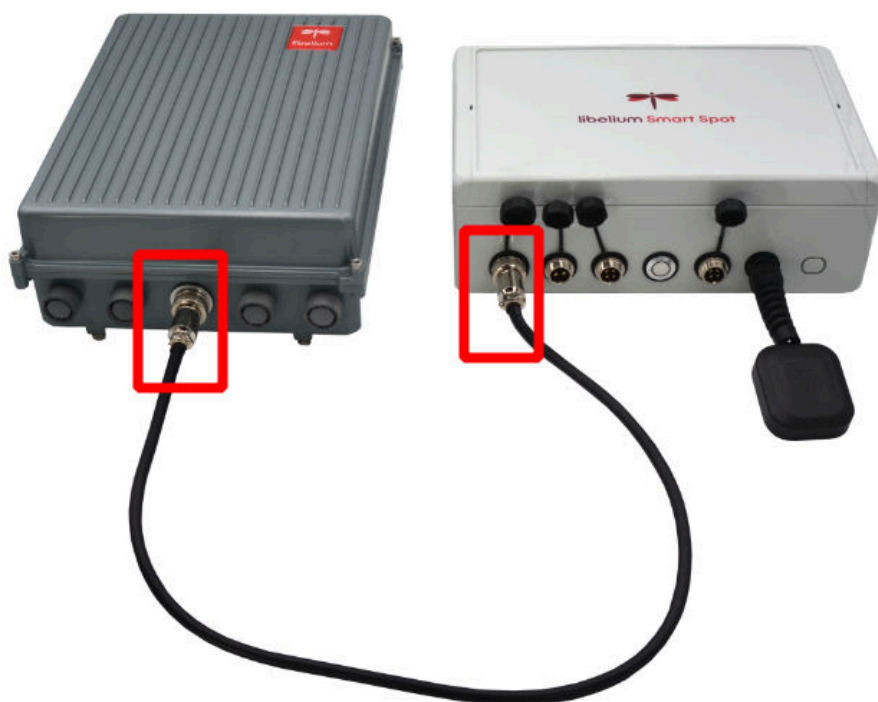


Figure: Visual inspection of connection between main and Air quality expansion enclosures

The device consists of two boxes. The primary box corresponds to the communications module, and the secondary box corresponds to the expansion module, which may include air quality, particulate matter or noise sensors. The modules are connected to each other using a specific communications cable. In the event of a substandard cable connection on any of the modules, there is a possibility of communication failure with the expansion module.

- Please ensure that the communications cable is connected and securely fastened into both modules using the metal connector. Ensure that there is no slack in the connector at either end.

If you have checked that both modules are correctly connected and you still cannot see any data, please contact technical support by writing to support@libelium.com and indicating the following in the subject line:
Incident Air Quality Enclosure Data <Customer_Name> <DEVICE_ID>

4. Appendix I: Sonometer calibration

4.1. Maintenance intervals

It is imperative that sound level meters are periodically maintained in order to ensure the accuracy of the data collected. This maintenance process involves recalibrating the meters to ensure they are operating at their optimal level. In the case of the sound level meter incorporated in the SmartSpot device, it is recommended to perform recalibration at least once every 12 months.

In addition to the recommended one-year interval, it is also advisable to carry out additional recalibrations if significant changes in measurement accuracy are observed, if the sound level meter has been subjected to extreme environmental conditions, or if it has suffered physical damage. More frequent recalibration can be beneficial in verifying and ensuring the quality of the data during critical moments or when equipment malfunction is suspected.

4.2. Required tools

4.2.1. iris360 platform

The Smart Spot device is always accompanied by access to a remote device management platform called iris360. This platform serves as the remote management point for all users and Smart Spot devices, regardless of their communication technology.

Should you encounter any issues accessing the iris360 platform, please contact Libelium via email at support@libelium.com.

4.2.2. Sound Level Calibrator tool

In order to carry out the calibration procedure, it is necessary to have a specific tool, commonly known as a 'Sound Level Calibrator'. Please note that this tool is not provided with the SmartSpot and must be purchased separately. The standard noise level for calibrators of sound level meters is typically 94 dB (decibels) at a frequency of 1 kHz (kilohertz).



Figure: Example of Sound Level Meter Calibrator

The sound level meter calibrator emits a sound signal at that specific level and frequency, and the sound level meter incorporated in the SmartSpot should accurately detect and measure this level. Should the sound level meter register a reading that differs from 94 dB at 1 kHz during calibration with the calibrator, adjustments must be made to correct any deviations and ensure precise measurements in future applications. The checking and adjustment process via the iris360 platform is outlined in the following sections.

4.3. Calibration procedure



The subsequent steps assume that the SmartSpot device is already switched on and configured. For further information, please refer to the Quick Start or Technical Guide.

The Smart Spot receives commands in response to data reporting. If the SmartSpot reporting interval is configured with a large interval (e.g. 5 min) and with other Device Mode different than 'Mode 0 (Continuous Mode)'; it is recommended to reduce it to the minimal possible value depending on the device hardware configuration and establish 'Mode 0 (Continuous Mode)' to facilitate the checking (e.g. 2 minute). Please refer to the Quick Start or Technical Guide if you require further information to carry out this procedure.

The following steps outline the procedure for calibrating the Smart Spot sonometer:

- **Step 1:** Configure the Sound Level Calibrator to 94dB.



94dB is a standard calibration level commonly available on the Sound Level Calibrators. The Smart Spot is configured by default to carry out the calibration at 94dB. Please note that 94dB is not mandatory, as the Smart Spot calibration level can be configured using iris360.

- **Step 2:** Log in to the Iris360 platform using your user name and password.
- **Step 3:** Navigate to the 'Data Sources' > 'Devices' section.
- **Step 4:** Use the filter box to search for the device(s) you wish to configure.
- **Step 5:** Select the device you wish to configure.



Figure: Navigation process through Iris360 to the device.

- **Step 6:** Within the 'Related Entity' section, select the entity with the suffix 'NLO'.

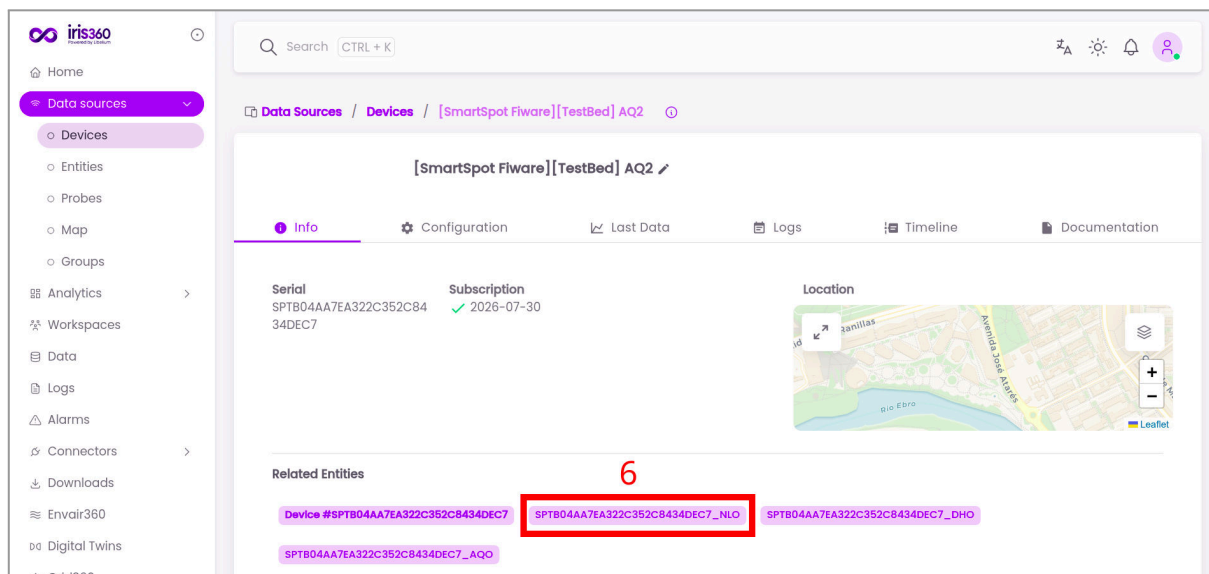


Figure: Location of the 'Noise Level Observed (NLO)' entity within the device.

- **Step 7:** Navigate to the 'Last Data' tab within the iris360 platform.
- **Step 8:** Verify that the 'Calibration reference' measure in iris360 is 94dB, which matches the Sound Level Calibrator configuration.



If the 'Calibration reference' configuration is not 94dB, it is possible to configure it via the 'Commands' tab.

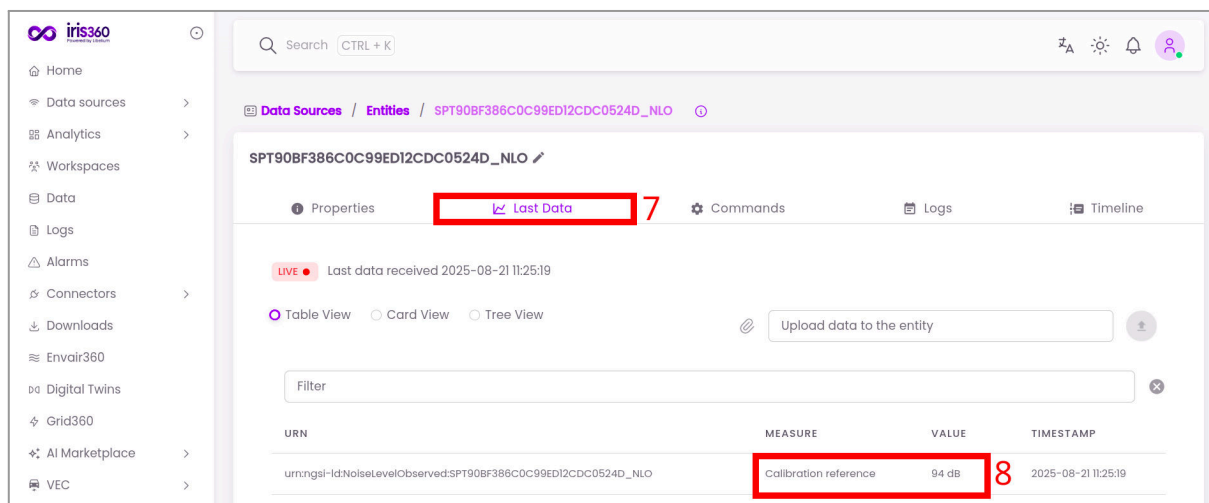


Figure: Location of the 'Last Data' tab within the 'Noise Level Observed (NLO)' entity.

- **Step 9:** Remove the SmartSpot sonometer windshield protector with care.
- **Step 10:** Ensure that the Sound Level Calibrator is correctly positioned within the SmartSpot sonometer, ensuring that the sonometer is fully inserted into the Sound Level Calibrator.



Figure: Sound Level Calibrator placed in the SmartSpot sonometer

- **Step 11:** Switch on the Sound Level Calibrator.
- **Step 12:** Navigate to the 'Commands' tab within the Iris360 platform.
- **Step 13:** Set the 'true' value within the 'Calibrate sonometer' command and click on 'Send' button to request a calibration procedure for the device.

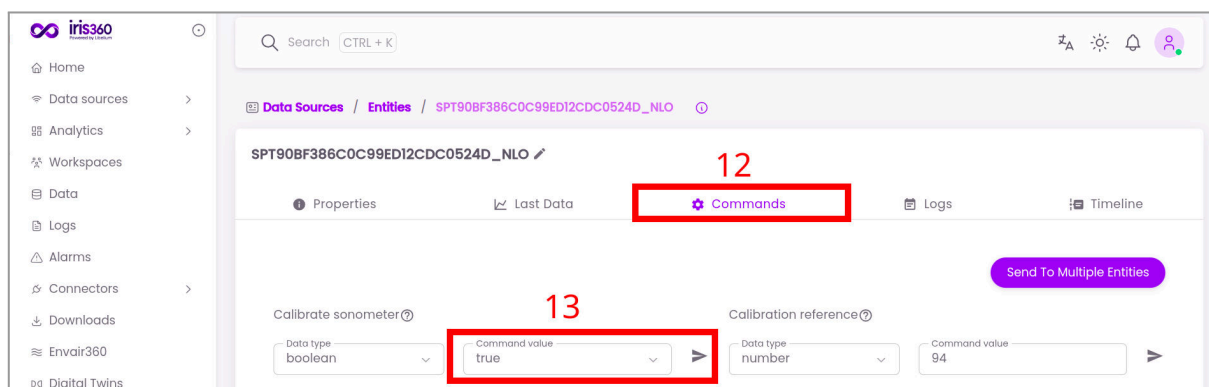


Figure: Location of the 'Command' tab within the 'Noise Level Observed (NLO)' entity.

- **Step 14:** Wait until the next measurement is reported by the device for the execution of the calibration.



Commands are received in the device as response to the value reporting procedure, so the calibration will be carried out during the next 2-3 seconds after receiving the report.

- **Step 15:** Wait for the next measurement, which must be an already calibrated measure. The value of 'LAeq' in 'Last Data' tab must be a value in between 93.8dB and 94.2dBm



Repeat the process from Step 13 if the value is not correct to ensure a correct calibration.

- **Step 16:** Following the completion of the procedure:
 - Switch off and remove the Sound Level Calibrator
 - The SmartSpot sonometer windshield protector should be installed with care, and the original Device Mode and reporting interval should be set up in case the device was modified during the calibration procedure.

5. Appendix II: Sonometer measurement compensation

In accordance with the ISO 19962:2009 standard, the unit **should be installed at a distance of 2 metres from a wall. Avoid installing the product on walls wherever possible.**

If this is not feasible, please follow the next steps in order to apply a compensation in the measurement to remove the reflected noise.

- **Step 1:** Log in to the Iris360 platform using your user name and password.
- **Step 2:** Navigate to the 'Data Sources' > 'Devices' section.
- **Step 3:** Use the filter box to search for the device(s) you wish to configure.
- **Step 4:** Select the device you wish to configure.



Figure: Navigation process through Iris360 to the device.

- **Step 5:** Within the 'Related Entity' section, select the entity with the suffix 'NLO'.

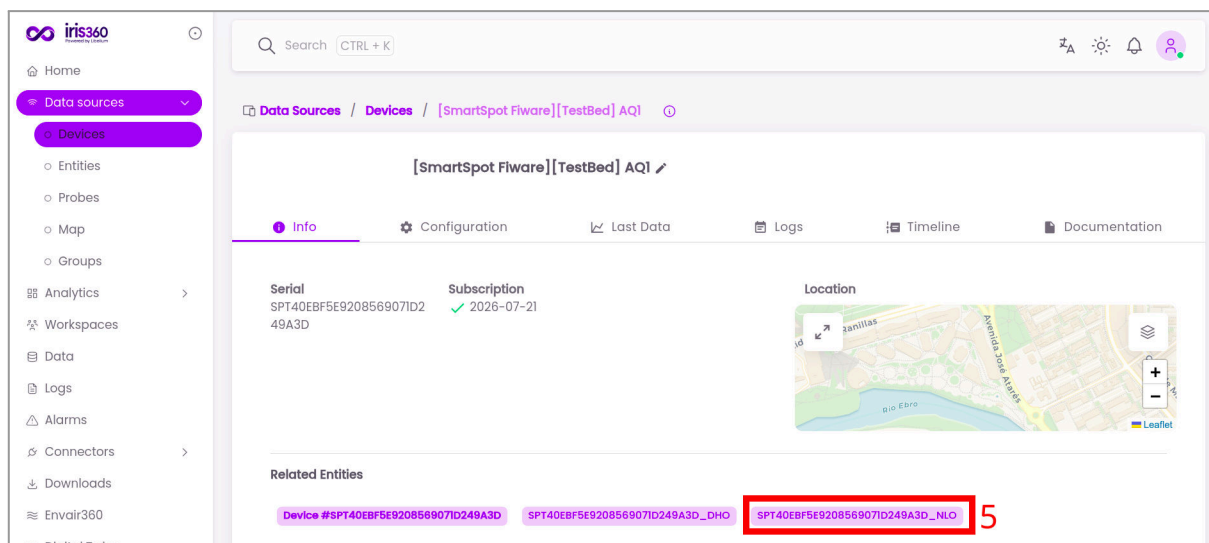


Figure: Location of the 'Noise Level Observed (NLO)' entity within the device.

- **Step 5:** Navigate to the 'Commands' tab within the Iris360 platform.

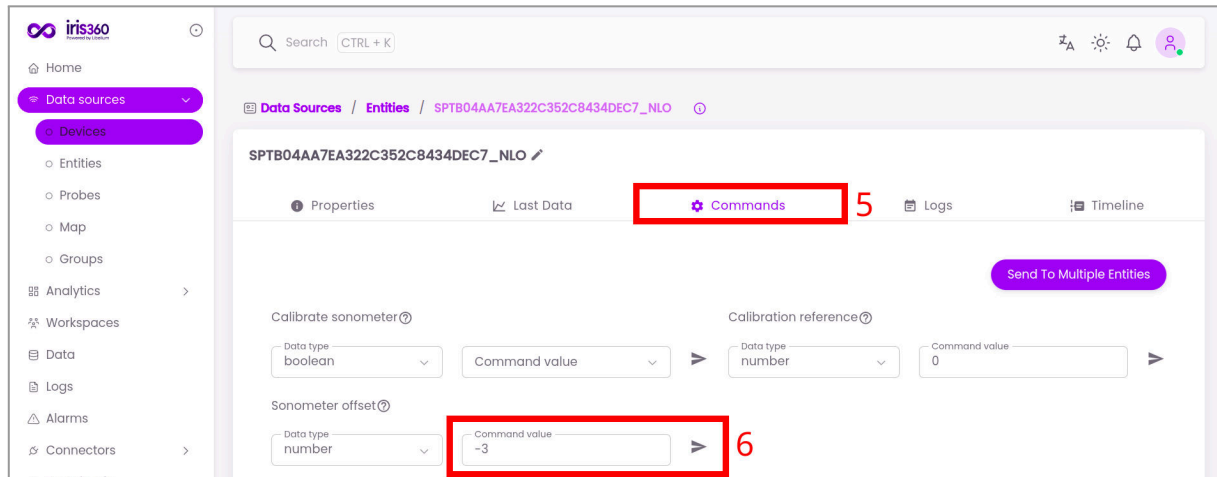


Figure: Location of the 'Command' within the 'Noise Level Observed (NLO)' entity.

- **Step 6:** Set the '-3' value within the 'Sonometer offset' command to apply the change on the device. This will be set up on the next report sent by the device.



The '-3' value is an approximate and recommended figure within the normatives. However, it is possible to calculate the specific value by following the established normative procedures.

6. Changelog

Version	Description	Date
v1.0	Document aligned to v1.3.0 firmware release First document version	10/10/2025

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